

Oluwatobi Melvyn Mayungbo

Data Scientist, Machine Learning and Statistics

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SUMMARY

Data scientist with a B.Sc. in Statistics who builds models for real operational use. Built forecasting with backtesting and cost-optimal decisions, sequence models over event data, and experiments with power analysis, and deployed services with Docker and FastAPI. Explains results clearly to non-technical users.

SKILLS

Statistics and Causal Inference: Time-Series Backtesting, Experiment Design, Power Analysis, Statistical Inference, CUPED, Causal Inference (S/T/X-Learners, Causal Forests), Uplift Modeling, Qini and AUUC, Logistic Regression, WOE Scorecards

Machine Learning: Scikit-learn, LightGBM, PyTorch, Feature Engineering, XGBoost, Deep Learning, Sequence Models (GRU, TCN, Transformers), Model Calibration (Isotonic Regression, Temperature Scaling), SHAP

Engineering and MLOps: Python, SQL, Docker, FastAPI, DuckDB, PostgreSQL, Redis, GCP Cloud Run, AWS ECS Fargate, Railway, Prefect, dbt-core, MLflow, Evidently, CI/CD (GitHub Actions), Git, Pytest, Streamlit, React

Data Analysis: Pandas, NumPy, SQL, Anomaly Detection

Payments and Risk: Transaction Fraud Detection, Credit Risk Decisioning, Payment Reconciliation, Double-Entry Ledgers and Idempotency, Regulatory Compliance (CBN/NDPA), Demand Forecasting, Web3 Fraud Risk Scoring

AI and LLM Systems: Retrieval-Augmented Generation (RAG), Hybrid Search (BM25, Dense Embeddings, RRF), LLM Evaluation and Guardrails, Prompt Engineering, Citation Verification, Model Context Protocol (MCP), Hugging Face, ONNX Runtime

PROJECTS

Self-Operating Demand Forecasting Platform | [GitHub](#)

Tech: Python, Prefect, dbt-core, DuckDB, LightGBM, MLflow, Evidently, FastAPI, Docker, GitHub Actions

- Built an end-to-end hourly demand forecasting system for NYC taxi zones, producing 24-hour quantile forecasts through an orchestrated Prefect pipeline spanning ingestion, dbt/DuckDB transformations, backtesting, drift monitoring and conditional retraining.
- Modelled asymmetric business cost by penalising unmet demand at 3× idle supply and shipping the cost-optimal quantile rather than the median, cutting expected cost by 25% on an identical model and beating a seasonal-naive baseline by 32%.
- Validated with rolling-origin backtesting across 6 expanding-window folds against seasonal-naive and historical-mean baselines, reporting MASE and pinball loss, with automated leakage tests that fail the build if any feature reads past the forecast origin.

Sequence-Based Transaction Fraud Detection System | [GitHub](#)

Tech: Python, PyTorch, LightGBM, ONNX Runtime, FastAPI, Docker, SHAP

- Built and compared three deep sequence architectures (GRU, Temporal Convolutional Network and a causally masked Transformer encoder) over customer transaction histories, with learned embeddings for high-cardinality merchant (~2.9k vocabulary), MCC and device-type categoricals; all models under 115k parameters and CPU-trainable in under 6 minutes.
- Deployed real-time scoring by exporting to ONNX Runtime (2.9× faster than eager PyTorch at batch size 1) and serving via FastAPI, reaching 5.1 ms p99 end-to-end latency against a 50 ms budget, with append-only per-decision audit logging and attention-based attribution of the prior transactions driving each flag.

Uplift Modelling for Campaign Targeting | [GitHub](#)

Tech: Python, scikit-learn, LightGBM, EconML, pytest, GitHub Actions

- Ran a causal inference study on the Hillstrom randomised e-mail trial (64,000 customers) to decide which customers to contact under a fixed budget, delivered as a two-page decision memo for a marketing director.
- Added a power and MDE calculator and a CUPED variance-reduction demo; 135 tests on synthetic data with CI on Python 3.10 to 3.12.

Premier League Match Predictor | [GitHub](#)

Tech: Python, LightGBM, scikit-learn, pandas, SQLite, Flask, Vercel

- Backtested gameweek by gameweek over 1,050 matches: 52.1% outcome accuracy against 43.2% for always picking the home side, and log loss 0.9948 against 1.0061 for an Elo-only baseline, with well calibrated probabilities.

EXPERIENCE

Machine Learning Instructor, SQI College of ICT

March 2026 to Present

- Teach machine learning and deep learning to student cohorts, covering statistical foundations, classical ML algorithms, neural architectures, sequence models and production evaluation metrics.
- Instruct students on data analysis, feature engineering and model building with PyTorch, Scikit-Learn, Pandas and NumPy.

Internal Auditor, NISER

May 2024 to July 2024

- Analysed financial and operational records to identify inconsistencies and irregularities, performing manual anomaly detection across complex transactional datasets.

EDUCATION

B.Sc. Statistics, University of Ibadan

January 2021 to March 2025

Professional Diploma in Artificial Intelligence, SQI College of ICT

July 2025 to Present

CERTIFICATIONS

- Google Data Analytics Professional Certificate
- Google Advanced Data Analytics Professional Certificate
- Claude Code In Action